TH series
Industrial Robot

MAINTENANCE MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.2
April 1, 2020

Shibaura Machine

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The information contained in this manual is subject to change without notice to effect improvements.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

! CAUTION

- Be sure to read through the safety instruction manual provided separately before starting actual work to assure the safe work covering from the robot installation to operation.

[Maintenance and inspection]

Be sure to observe the following items to use the robot safely.

! DANGER	
 Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
 Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

! CAUTION	
 Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

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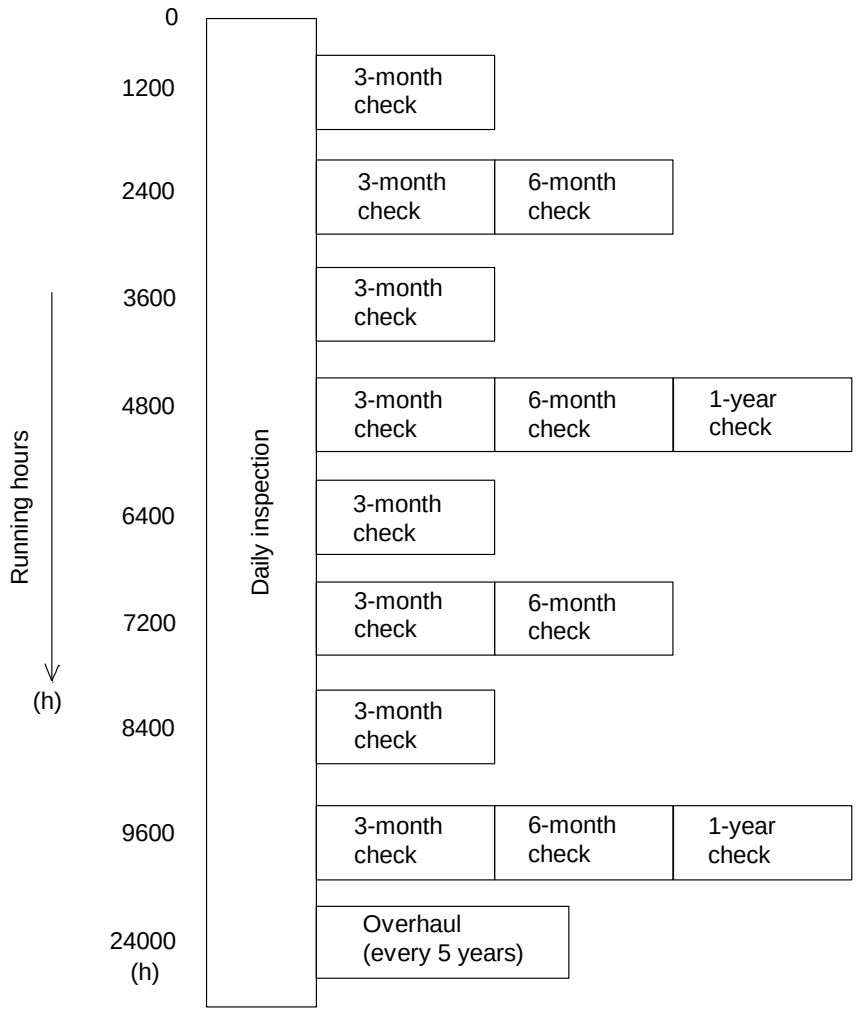
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Section 1 Maintenance Items

1.1 Maintenance Schedule

Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule

- When the system is operated for 16 hours per day:
  16 hours × 25 days × 3 months = 1,200 hours
- When the system is operated for 24 hours per day:
  24 hours × 25 days × 3 months = 1,800 hours

1.2 Items for Maintenance and Inspection

This section describes the items for maintenance and inspection.
For the executing procedures, see the relevant paragraph listed in the table below.

a) Daily inspection

No.	Item	Inspection procedures	Relevant Para.
1	Cable ducts	(1) Make sure that any cable duct will not twist, bend or crack locally at the time of robot motion. (2) Make sure that all connectors of the cable ducts are tightened completely.	2.3.1
2	Hand set flange	Make sure that the hand set flange bolts are tightened completely.	2.3.2
	Robot installation bolts	Make sure that the robot base installation bolts are tightened completely.	
3	Vibration and noise	Make sure that each axis can move smoothly.	
4	Change in positioning accuracy	Make sure that the robot can stop at a predetermined position accurately.	
5	Air leakage (When air piping is used)	Make sure that the air will not leak from the robot interior or air joints.	2.3.3
6	Controller vent holes	(1) Make sure that no obstacle stands in front of the vent holes. (2) Make sure that no contaminant is left in the vent hole.	3.3.1
7	Function of safety devices	Make sure that each safety device connected to the emergency stop pushbutton switch and external operation input signal can function properly.	3.3.2

b) Quarterly (every three (3)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Lubricated state of ball screw spline	Make sure that the ball screw spline (or tool shaft) is greased completely and that no rust is developed on it.	2.3.4

c) Semi-annual (every six (6)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Greasing of ball screw spline	Apply the grease to the ball screw spline.	2.3.4
2	Each axis drive system	Move each axis by hand and make sure that it moves smoothly.	2.3.5
3	Timing belts	(1) Make sure that each timing belt is not worn off or cracked. (2) Check each timing belt for tension.	2.3.6

d) Annual (every year) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Lubrication of ball screw	Fill grease to the ball screw spline.	2.3.4
2	Lubrication of reduction gears	Fill grease to the reduction gears from the grease nipples.	2.3.7
3	Wiring and piping	Make sure that all cables and air tubes are in the good state.	
3	Replacement of position detector batteries (Be sure to replace the batteries every year.)	Replace the position detector backup batteries (housed in the robot base unit). If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage drops, a battery alarm occurs. Then replace the batteries immediately.	2.3.8

e) Overhaul (every five (5) years)

No.	Item	Inspection procedures	Relevant Para.
1	Inspection of consumable parts	Inspect the flexible cables, reduction gears, ball screw splines, timing belts and motors, then replace them when necessary.	
2	Battery replacement (Replace the battery every five (5) years.)	Replace the RAM (controller) backup battery. If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage drops, the battery alarm message saying “1–145 MAIN Battery alarm” is displayed. Then replace the battery immediately.	3.3.3
3	Replacement of switching power supply unit (Replace the battery every five (5) years.)	As the switching power supply unit uses an aluminum electrolytic condenser, its life is limited. Replace the switching power supply unit (DC5 V, DC24 V) for preventive maintenance. The life of the switching power supply unit differs with load current, running hours and ambient temperature. If the load current is large, the running time is long or the ambient temperature is high, the life becomes short.	3.3.4

1.3 Maintenance Tools

We recommend to use the following tools for maintaining the robot and controller.

- Phillips screwdrivers
- Hexagonal wrench set: M3 ~ M8
- Assembly wrench set
- Water pump pliers
- Nippers
- Grease guns

1.4 Maintenance Contract and Repair

1.4.1 Maintenance Contract

With the exception of greasing and lubrication, we recommend that the user conclude with us an after-sale service contract on the inspection items for maintenance and inspection to be done semi-annually and after.

1.4.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller.

- Robot model
- Manufacture number
- Date of manufacture
- Operating time

1.4.3 Modification

This robot and controller MUST NOT be modified without the consent from SHIBAURA MACHINE.



CAUTION

- The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.

Section 2 Maintenance of the Robot

2.1 Cautions on Maintenance and Inspection

When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

! DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

2.2 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 2.1 and Fig. 2.2.
The drive mechanism of each axis is described below.

a) Axis 1
The output of axis 1 AC servo motor is input to the reduction gear. The output of the same gears directly controls the arm 1 motion.

b) Axis 2
The output of axis 2 AC servo motor is input to the reduction gear. The output of the same gear directly controls the arm 2 motion.

c) Axis 3
The output of axis 3 AC servo motor is input to the ball screw nut through the timing belt to move the ball screw spline up and down.

d) Axis 4
The rotation of axis 4 AC servo motor is input to the reduction gear and the output of axis 4 AC servo motor is input to the ball spline nut through the timing belt to revolve the ball screw spline shaft.

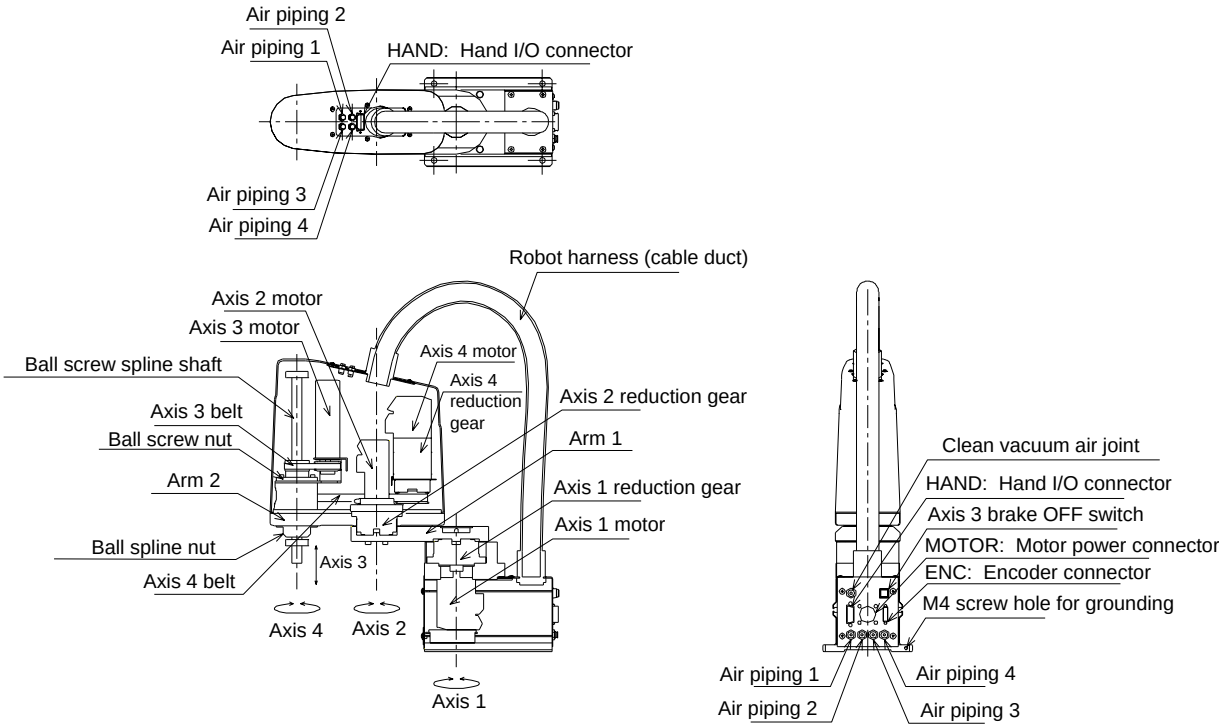


Fig. 2.1 Layout of robot mechanical components (TH250A, TH350A)

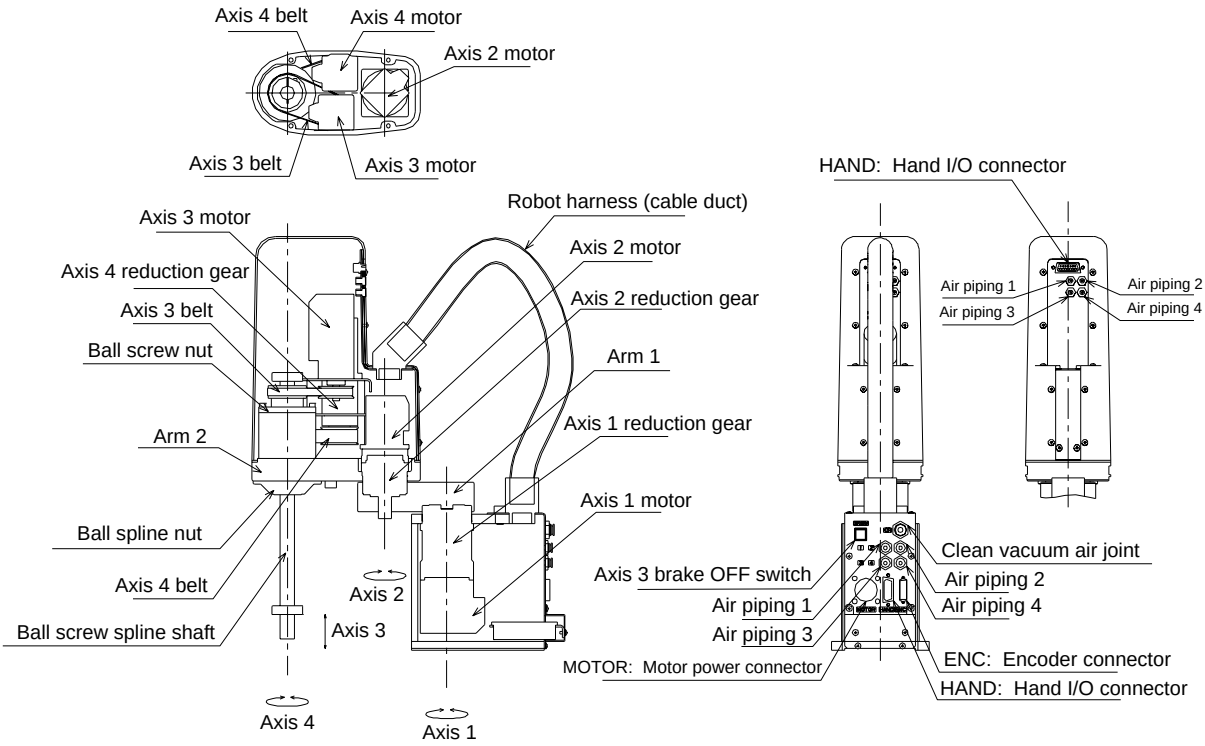


Fig. 2.2 Layout of robot mechanical components (TH180)

2.3 Robot Maintenance Procedures

2.3.1 Check of Cable Duct for Abnormality

a) Visually check to see that the cable duct is not locally twisted, bent or cracked when the robot moves. If there is any abnormality, replace the cable to prevent breakage of the cable wire, etc.

b) The connectors securing the cable duct are secured by tightening it to the plate with a hexagon nut. If the hexagon nut is loosened, tighten it with a pair of water pump pliers, etc.

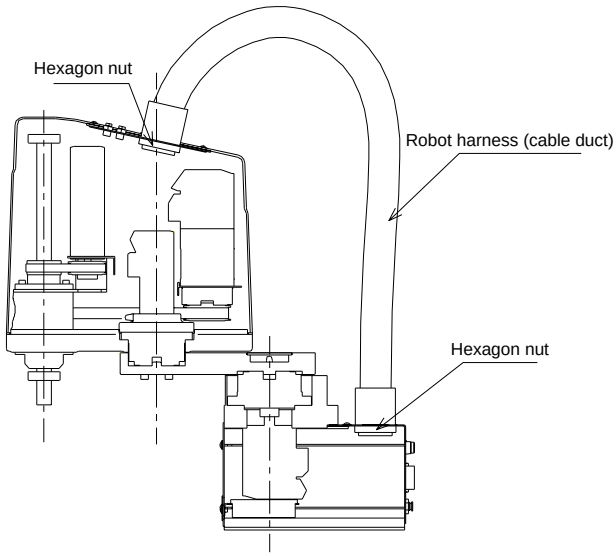


Fig. 2.3 Cable layout (TH250A, TH350A)

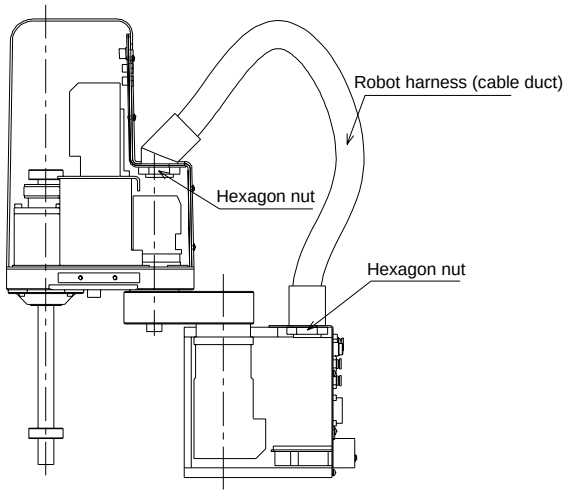


Fig. 2.4 Cable layout (TH180)

2.3.2 Check of Hand and Base for Mounting

DANGER

- Completely tighten the hand clamp bolt. Otherwise, the hand may drop or shift from a predetermined position.

a) Check of hand set flange (option) for looseness

Make sure, using a hexagonal wrench key, that the hexagon socket head cap screw of the hand set flange that is tightened to the ball screw spline is not loosened. Otherwise, tighten it.

Also make sure by means of a hexagonal wrench key that the hexagon socket head cap screw securing the hand set flange and hand is not loosened. Otherwise, tighten it.

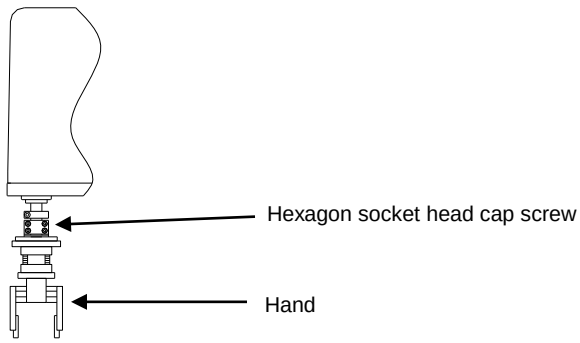


Fig. 2.5 Hand set area

b) Check of basic robot installation bolts

Make sure, using a hexagonal wrench key, that the installation bolts of the robot base are not loosened. Otherwise, tighten them.

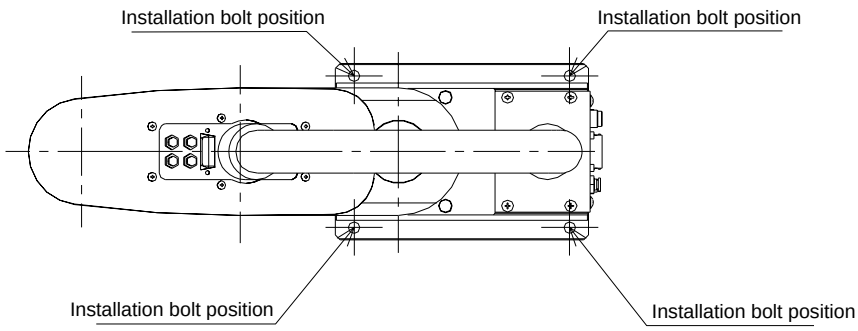


Fig. 2.6 Installation bolt positions

2.3.3 Check of Air Leakage

Piping connections in the robot are shown in Figs. 2.7 and 2.8. Make sure that no air leaks from the robot interior or air joints.

DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

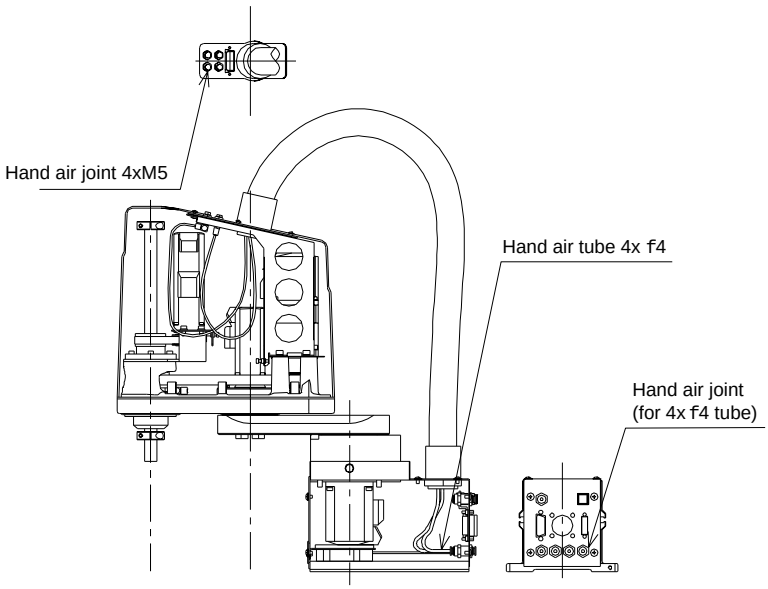


Fig. 2.7 Hand air piping (TH250A, TH350A)

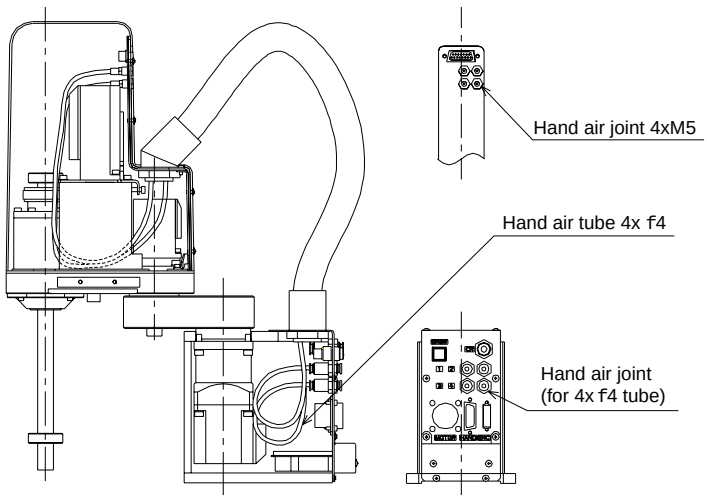


Fig. 2.8 Hand air piping (TH180)

2.3.4 Lubrication of Ball Screw Spline Shaft

To lubricate the ball screw spline shaft, use the following grease.

- FOMBLIN grease (Ex. OT–20 made by Solvay Solexis)

! CAUTION

- DO NOT mistake the type of grease. Otherwise, the robot performance may deteriorate or the robot may malfunction.
- After you have touched the ball screw spline by hand, be sure to apply the grease to it. Otherwise, rust will develop on the shaft.

a) Check of ball screw spline shaft for lubrication
Open the arm 2 cover and make sure that the ball screw spline shaft has not run out of grease and that no rust has developed on it. If the ball screw spline shaft has run out of grease, apply the grease according to Para. b).

b) Greasing of ball screw spline shaft
Apply the grease with a brush or by hand so that all guide slots around the shaft are covered. Just after the greasing, DO NOT start an operation. First, move the ball screw spline shaft slowly over the entire motion range. Then wipe off any grease that oozes from the shaft.
Repeat this operation a few times.

2.3.5 Check of Each Axis Drive Status

Move each axis by hand to make sure that it can move smoothly.
For the axes 3 and 4, make them in the servo-free status and make sure of the above.

! DANGER

- When nearing the robot in the servo-free status, be sure to carry out the work by two (2) persons. One person should carry out the work while the other is monitoring the work outside the hazardous zone. The latter person should watch the work and be ready to press the EMERGENCY stop switch if any abnormality occurs.

2.3.6 Check of Timing Belt Status

Remove the arm 2 cover, and visually check to see that the belts of axis 3 and axis 4 are not worn off or cracked.
Also check the timing belt for tension and adjust as necessary, referring to Table 2.1 below.

Table 2.1

Robot model	Axis	Tension [N]	Unit mass [g/cm^2]	Belt width [mm]	Span [mm]
TH250A	3	48 ~ 64	0.22	9	49.5
TH350A	4	66 ~ 88	0.22	12	183
TH180	3	44 ~ 50	0.025	9	59
	4	96 ~ 110	0.040	15	63

2.3.7 Greasing of Reduction Gears (TH250A, TH350A)

To lubricate the reduction gears, use the following grease.

- Harmonic grease 4B No.2 (Maker: Harmonic Drive Systems)

! CAUTION

- DO NOT mistake the type of grease. Otherwise, the robot performance may deteriorate or the robot may malfunction.

Remove the axis 1 cover and arm 2 cover, and fill the specified volume of grease from the grease nipples, using a grease gun.
The greasing volume is 5 cm^3 and the greasing locations are shown in Fig. 2.9.
(For dismantling the arm 2 cover, see Para. 2.3.9.)
The type of grease nipple is A–M6F.
As the axis 4 reduction gear of TH250A and TH350A and axes 1, 2, 4 reduction gears of TH180 are grease-sealed, they need not be greased at all.

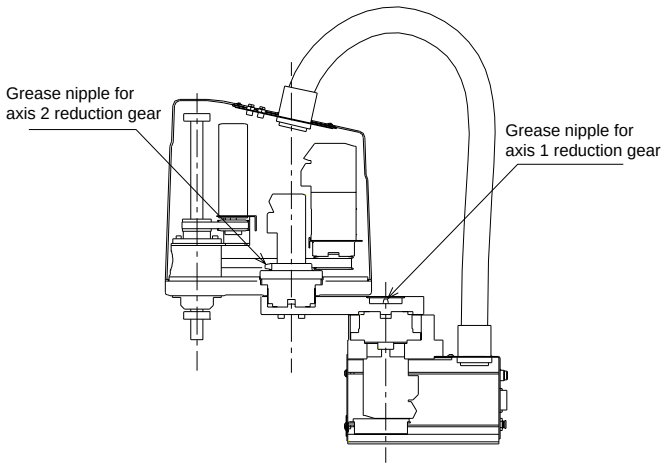


Fig. 2.9 Greasing locations of reduction gears (TH250A, TH350A)

! CAUTION

- After all greases in the axis 1 and axis 2 reduction gears have been replaced, be sure to perform running-in for more than half an hour at 5 % speed override. Otherwise, the performance of the reduction gears may deteriorate or they may malfunction.

2.3.8 Replacement of Position Detector (or Encoder) Batteries

To back up the data of the position detector attached to each motor, batteries are used.
The battery life changes with the controller power OFF time. When the controller is operated at a rate of eight (8) hours per day, the calculated life is approximately two (2) years which may vary, however, with the operating condition, ambient temperature, etc. Be sure to replace the batteries, therefore, at the time of yearly inspection. For the replacing procedures, see the descriptions below.

Note: The battery alarm comes in the two (2) types; battery warning alarm (Battery alarm) and battery error (Encoder abnormal).
Normally, when the battery warning alarm has generated, the battery error can be avoided by replacing the batteries with new ones. When this happens, however, the battery error (Encoder abnormal) will occur within one (1) month after generation of the battery warning alarm.
If the time of battery replacement is delayed, the battery error will generate and the robot may not be moved at all.
Once the battery error has been detected, the robot enters an emergency stop state, and reset of the encoder error is required.
(For how to reset the encoder error, see the Appendix B of this manual.
Be sure to replace the batteries at the time of annual inspection.

The following messages are displayed at the time of battery error.

8–065 Axis1 Encoder abnormal
8–097 Axis2 Encoder abnormal
8–129 Axis3 Encoder abnormal
8–161 Axis4 Encoder abnormal
8–193 Axis5 Encoder abnormal

The battery error can be identified when the following alarm has generated, which is displayed also in the error history at the same time.

1–049 Axis1 Battery alarm
1–065 Axis2 Battery alarm
1–081 Axis3 Battery alarm
1–097 Axis4 Battery alarm
1–113 Axis5 Battery alarm

The batteries should be replaced while the power is turned on and emergency stop is effected only after the safety is assured.

[Replacing procedures]

1) Assure the safety, then press the EMERGENCY stop switch to emergency-stop the robot while the power is turned on.
2) Remove the truss head screws 4×6 SUS or truss head screws 3×6 SUS from the robot base unit, and disengage the battery case or battery box. (See Fig. 2.10 (a) or Fig. 2.11 (b).)
3) Remove the batteries which are connected by means of the connector.
4) Set the new batteries.
5) Attach the battery case or battery box.
6) Reset the error through the teach pendant.

CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

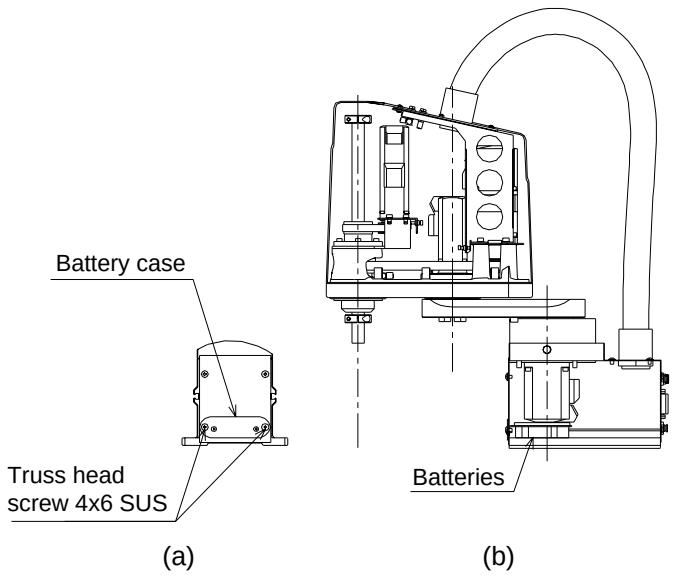


Fig. 2.10 Position detector battery set position (TH250A, TH350A)

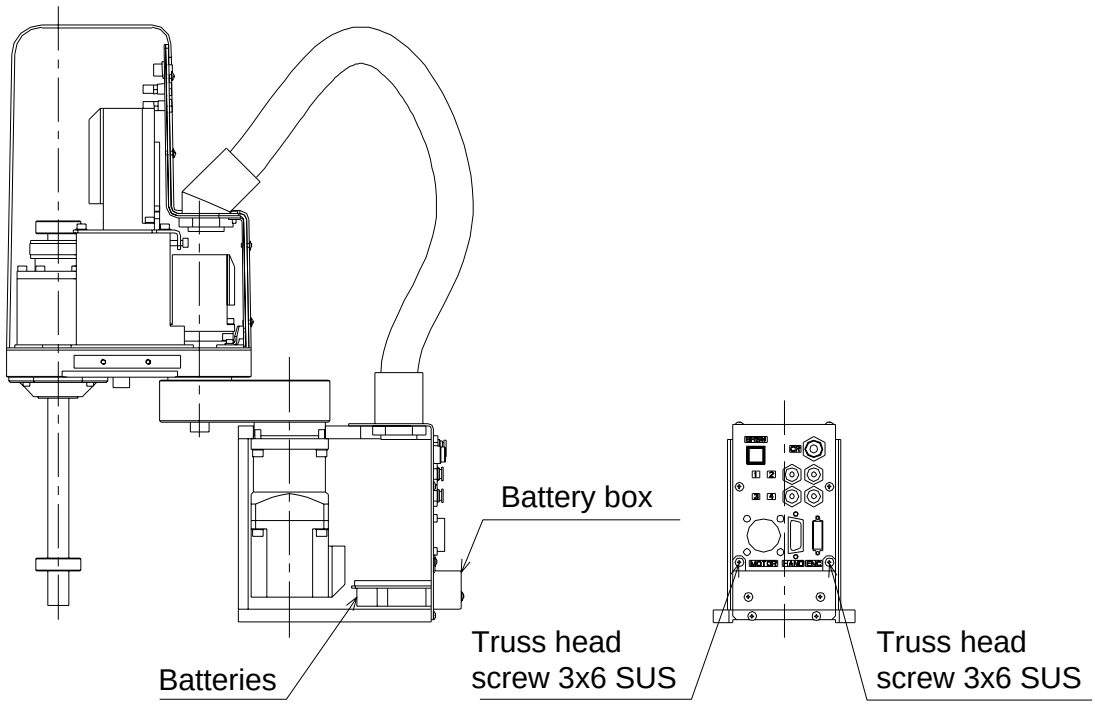


Fig. 2.11 Position detector battery set position (TH180)

2.3.9 Disconnection of Arm 2 Cover

First of all, provide the following tools.

- Hexagonal wrench key
- Plus driver

Turn off the controller power before the operation.
Remove the truss head screws 3×8 SUS and hexagon socket head cap screws 3×16 SUS as shown in Fig. 2.12, or the truss head screws 3×6 SUS and hexagon socket head cap screws 3×10 SUS as shown in Fig. 2.13.
Then pull the arm 2 cover upward and disconnect it.

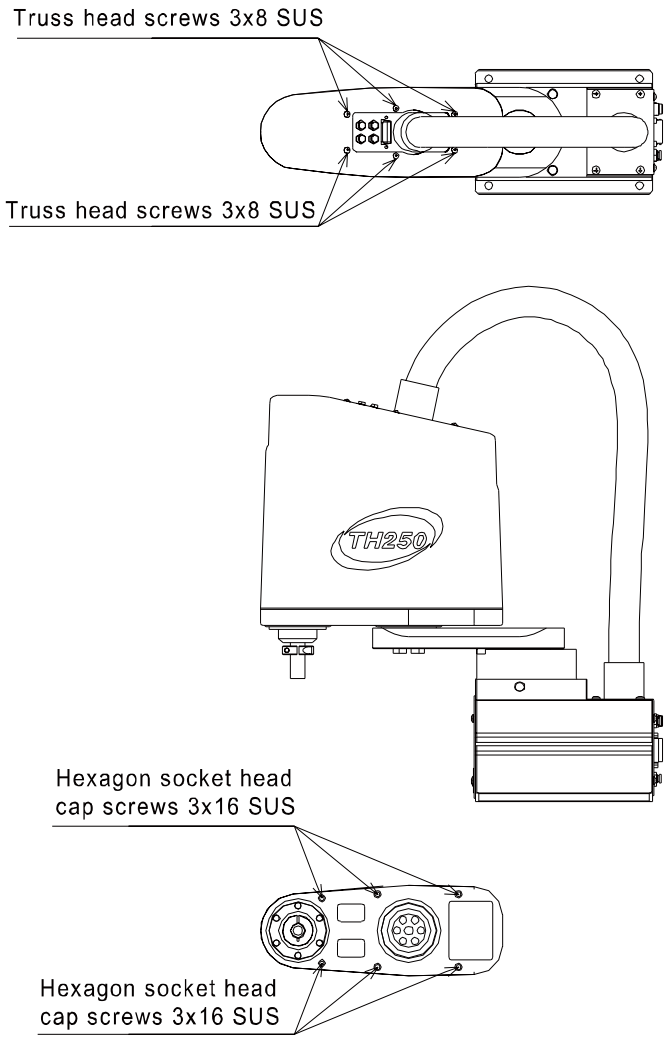


Fig. 2.12 Disengaging Arm cover (TH250A, TH350A)

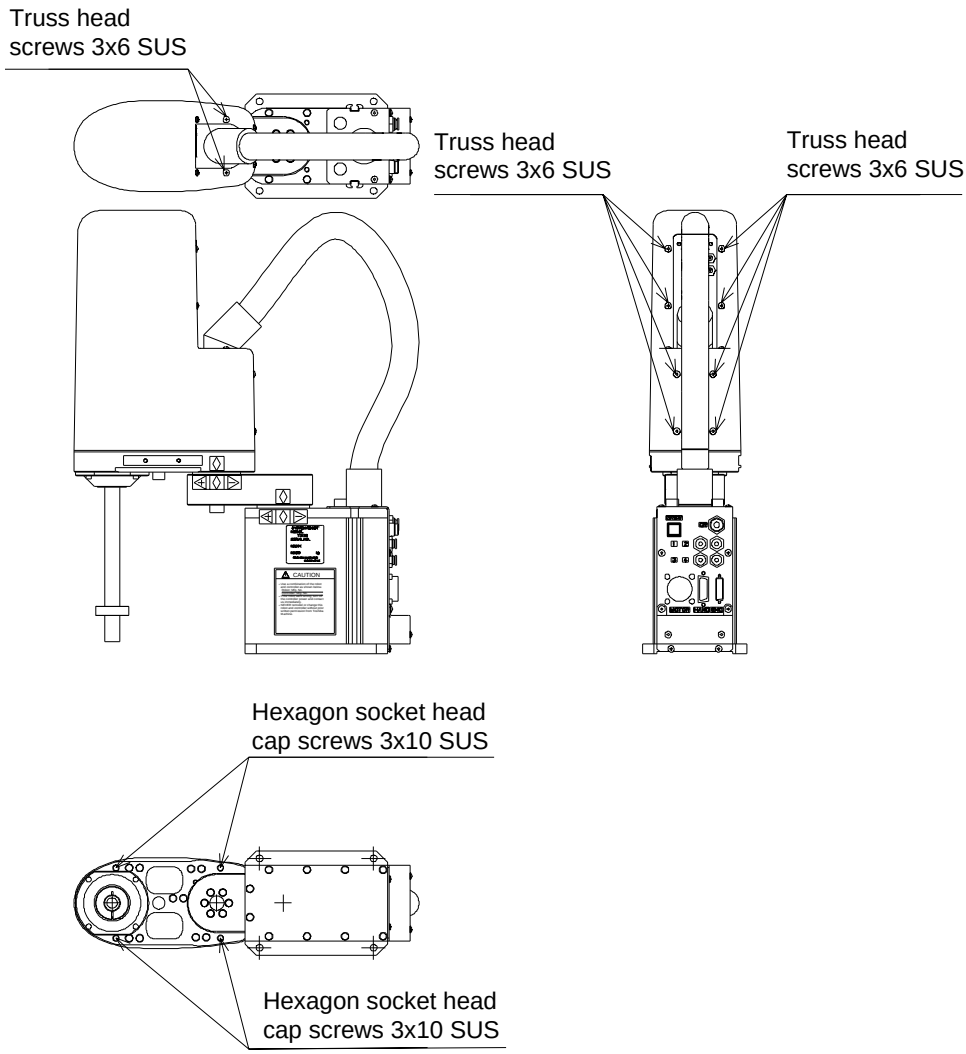


Fig. 2.13 Disengaging Arm cover (TH180A)

2.4 Cautions on Robot Transport and Storage

When transporting or storing the robot, it is recommended to secure the robot in the shipment posture, using the attached clamp for transport.

Section 3 Maintenance of Controller

3.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of robot controller TS1000 (hereinafter called the “controller”), follow the items given below so that the work can be carried out safely.

! CAUTION

- Before removing the controller cover for maintenance or inspection, be sure to turn off the main power switch of the controller.
  Also, turn off the power breaker when connecting or disconnecting the power cable.
- DO NOT touch the servo driver for at least two (2) minutes after the main power switch is turned off. The large-capacity capacitor in the servo driver may be charged with voltage, and touching the servo driver could lead to electric shocks.
- DO NOT disconnect the battery connector at other than battery replacement. Otherwise, files saved in the memory may be lost.

! DANGER

Be sure to turn off the main power switch. If the main power switch remains ON, the servo power printed board, servo printed board and switching power supply are charged, and touching them could lead to electric shocks.

3.2 Layout of Controller Parts

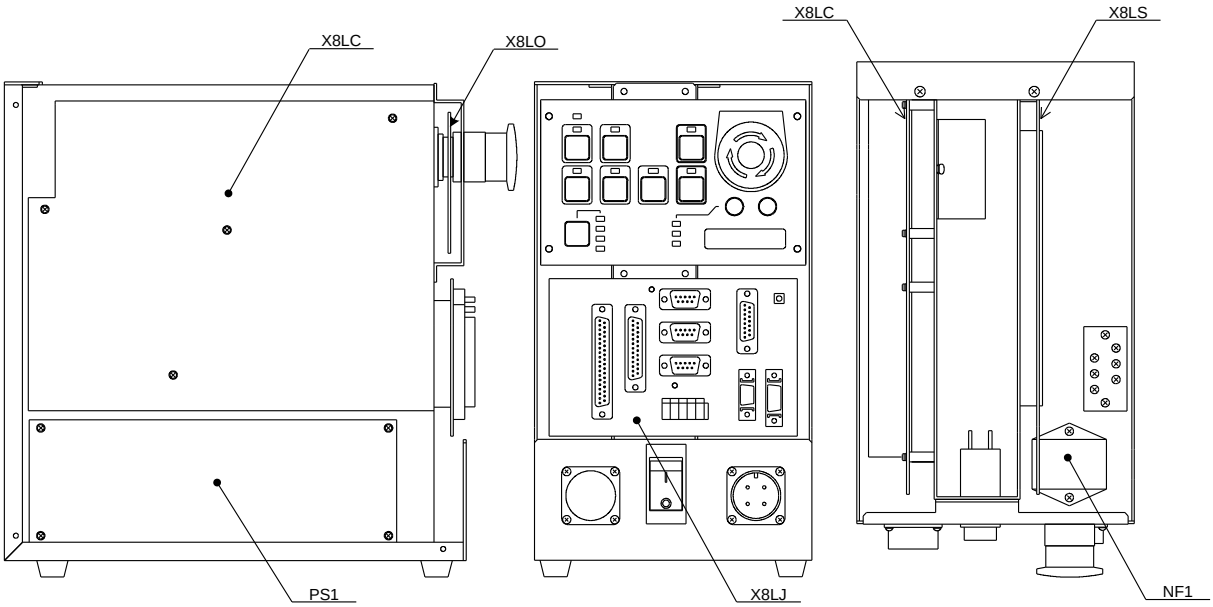


Fig. 3.1 Layout of controller parts

Part name	Descriptions
PS1	DC5V/DC24V switching power supply
X8LC	Main control printed board
X8LS	Servo power printed board
X8LJ	Mother board also serving as connector board
X8LO	Control panel printed board
NF1	Noise filter

3.3 Maintenance Procedures

3.3.1 Check of Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction.
To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.

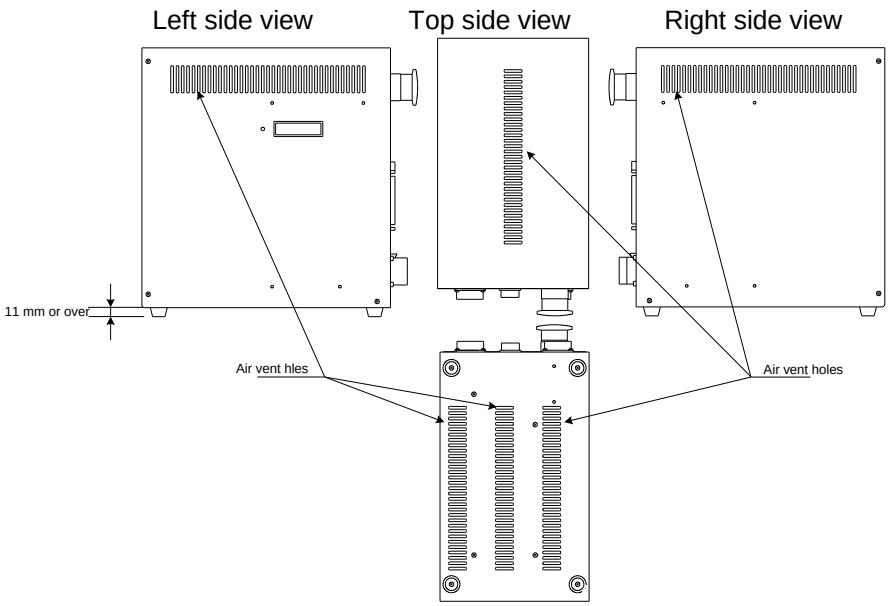


Fig. 3.2 Controller side views

a) Make sure that no obstacle stands in front of the air vent holes.
If there is any obstacle, move it away from the air vent hole so that air flow is not blocked.
Keep the spaces of 50 mm or more in the right and left directions and 100 mm or more in the upward direction.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is let on the vent hole, remove it.

c) If the controller is placed sideways, make sure that the air vent holes on the bottom surface of the controller are more than 11 mm away from the floor surface. Always keep the space of 11 mm or over. The length of rubber shoe is approximately 11 mm.

3.3.2 Check of Safety Devices for Function

Make sure that the emergency stop pushbutton switches equipped on the control panel and teach pendant TP1000 (hereinafter referred to as the “teach pendant”) work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

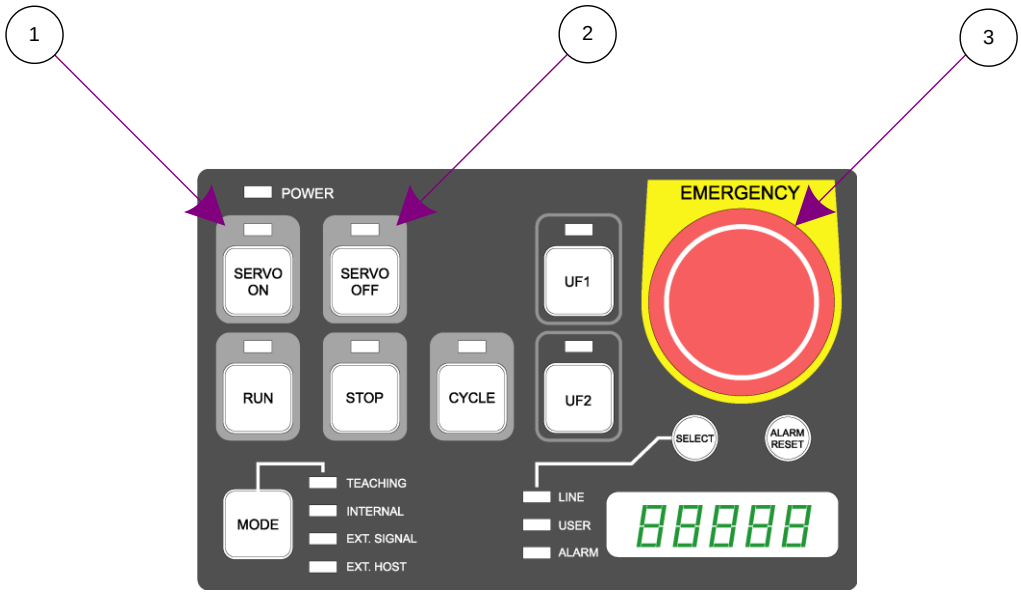


Fig. 3.3 Layout of control panel switches

a) Make sure that the EMERGENCY (emergency stop) switch [3] provided on the control panel functions properly.

1) Press the SERVO ON switch [1] provided on the control panel to turn the servos on and make sure that the SERVO ON switch [1] LED is illuminated.

2) Press the EMERGENCY switch [3] and make sure that the servos are turned off. The SERVO ON switch [1] LED goes out and the SERVO OFF switch [2] LED is illuminated. At this time, make sure that the EMERGENCY switch [3] remains depressed.

3) Turn the EMERGENCY switch [3] clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY STOP switch [5] provided on the teach pendant functions properly.

1) Press the SERVO ON switch [4] on the teach pendant to turn on the servo power, and make sure that the SERVO ON switch LED [4] is illuminated.

Note: The servo power can be turned on only in the TEACHING mode.

2) Press the EMERGENCY STOP switch [5] on the teach pendant and make sure that the servos are turned off. The SERVO ON switch LED [4] goes out.



Fig. 3.4 EMERGENCY STOP switch provided on teach pendant

At this time, make sure that the EMERGENCY STOP switch [5] remains depressed.

3) Turn the EMERGENCY STOP switch [6] clockwise and make sure that the same switch turns off.

c) Make sure that the safety devices controlled by the external operation signals work properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal “emergency stop” (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal “low speed command”, and make sure that each signal functions properly.

3.3.3 Battery Replacement

SRAM equipped on the X8LC printed board of the TS1000 robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the SRAM data, resulting in data such as user-set programs, parameters and absolute coordinate values being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

- 1–145 MAIN Battery alarm

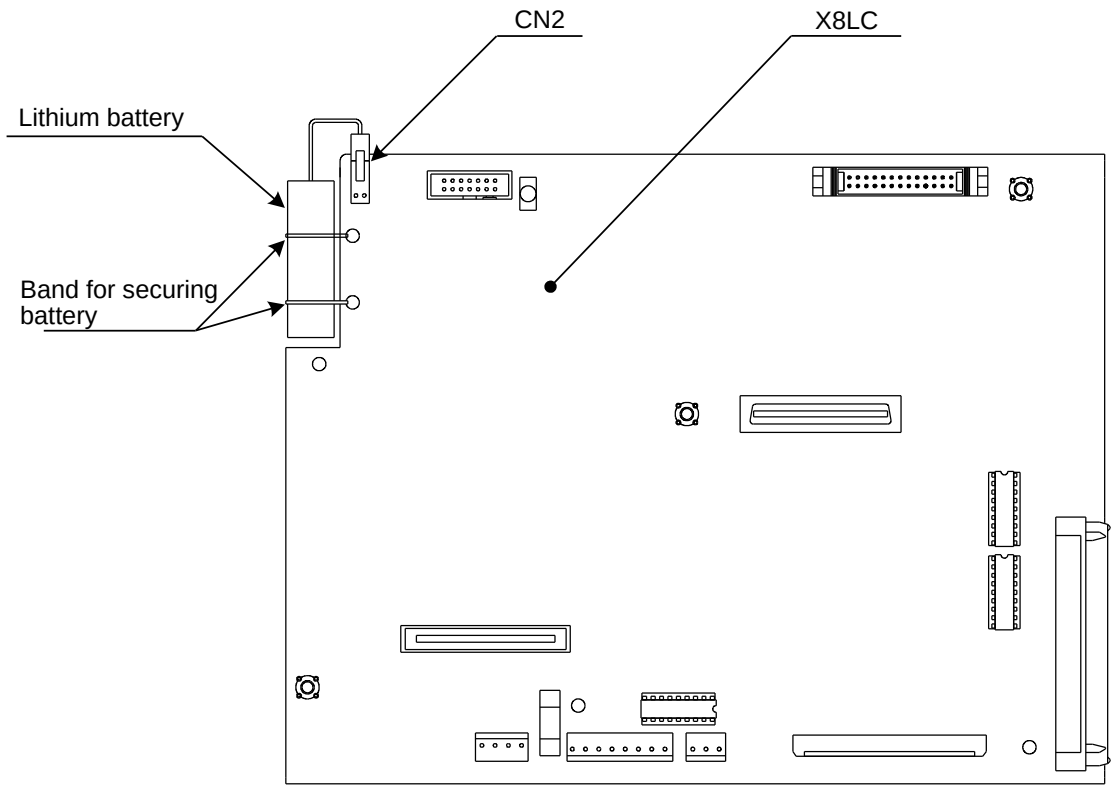


Fig. 3.5 Location of lithium battery

Table 3.2 Type of battery

Type of battery	Manufacturer	Battery service life
ER6V–C	TOSHIBA LIFESTYLE PRODUCTS & SERVICES Corp.	10 years (at room temperature)

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.

Only the battery shown in Table 3.2 should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) If the battery is to be kept removed for more than five (5) minutes, copy all programs and various parameters stored in the internal memory to the personal computer, using the HOST connector. In the worst case, all SRAM data are lost.

2) Set OFF the main power switch.

3) Remove the cover from the body.

4) Remove the battery connector connected with the X8LC printed board from CN2.
(For the X8LC printed board position, see Fig. 3.1.)

5) Remove the battery from the band securing the battery.
(For the battery position, see Fig. 3.5.)

6) Disconnect the connector, secure the new battery within five (5) minutes with the band, then connect the battery connector with CN2.
Even if not used, the lithium battery capacity decreases by natural discharge. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

7) Mount the cover on the body.

8) If the battery is kept removed for more than five (5) minutes, turn on the main power switch and load the programs and various parameters from the personal computer.

If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

! CAUTION

- Waste battery should be disposed of according to the user's in-house regulations.
- NEVER drop the battery into fire. NEVER short-circuit, charge, disassemble or heat it. Otherwise, liquid leakage or rupture may be caused.

3.3.4 Replacement of Switching Power Supply Unit

The life of the switching power supply unit (DC5 V/DC24 V) used in the TS1000 controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[Replacement procedures]

1) Turn off the main power switch.
2) Turn off the power breaker.
3) Remove the cover from the body.
4) Disconnect all cables which run from the power module to the X8LC, X8LS printed boards from CN1 and CN2.
   (For the position of each printed board, see Fig. 3.1.)
5) Disconnect the power module (LEB100F–0524 made by Cosel) from the body.
   (In all, four (4) screws are used.)
6) Mount a new power module and secure it with the four (4) screws.
7) Connect the cables which linked with CN1 and CN2, as originally connected.
8) Mount the cover on the body.

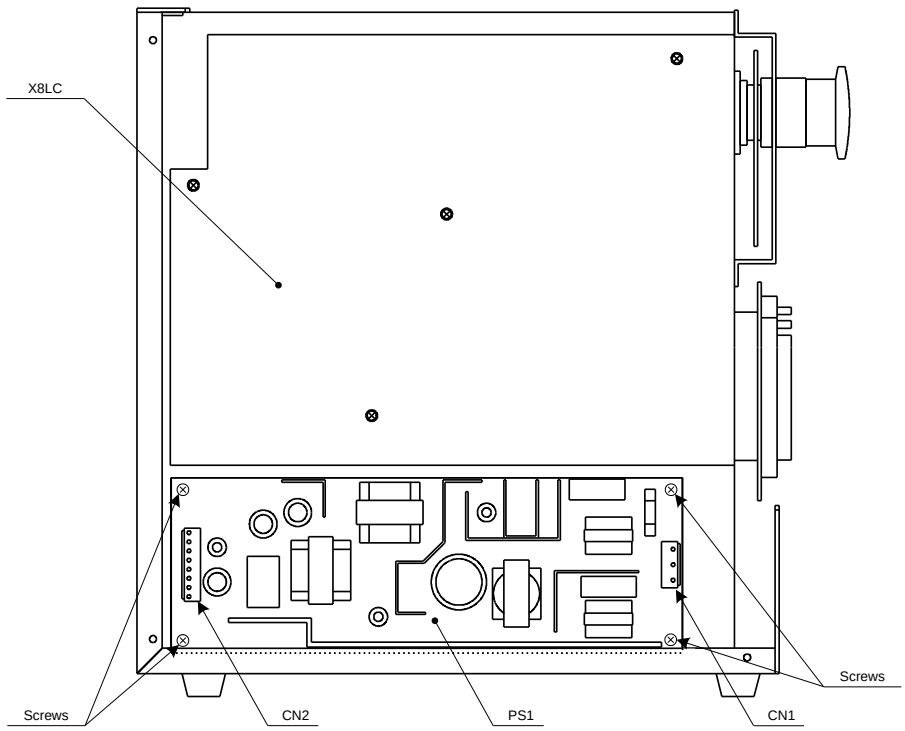


Fig. 3.6

3.3.5 Replacement of Fuse (X8LC Printed Board)

If the current exceeding the specified current has run through the I/O unit, the fuse of the X8LC printed board is blown out. If the alarm saying “X8LC Fuse broken. (or P24V power is abnormal.)” is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.

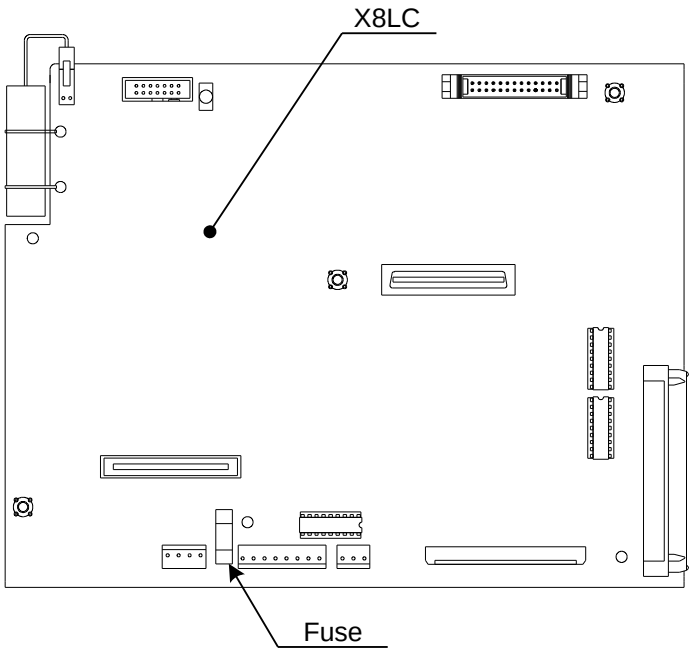


Fig. 3.7 Fuse layout

Table 3.3 Type of fuse

Type of fuse	Manufacturer
OMF–51NR 250 V 3A	World

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the cover attached to the fuse unit as shown above.
3) Remove the fuse and set a new fuse.
4) Mount the removed cover.
5) Turn on the controller main power and make sure that no error will occur.

3.3.6 Replacement of Output ICs (X8LC Printed Board)

If the current exceeding the specified current has run through the output unit due to wrong wiring or short-circuit, the outputs may not turn on normally.
When this happens, the fuse may be blown out and the X8LC printed board may be damaged. Replace the ICs then.
Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.

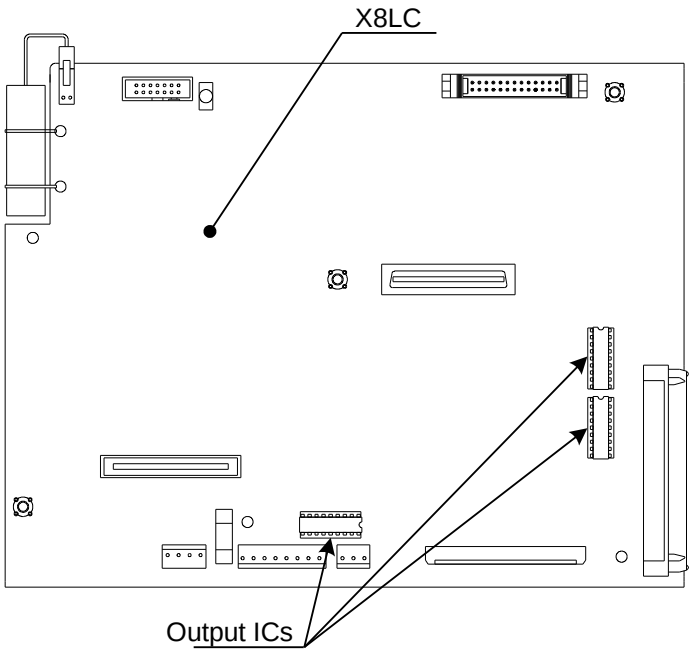


Fig. 3.8 Output IC layout

Table 3.4 Type of IC

Type of IC	Manufacturer
M54562P	Mitsubishi Electric

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the ICs shown above from respective sockets.
3) Attach new ICs to the sockets.
4) Turn on the controller main power.

Appendix Table 1 List of Mechanical Replacement Parts for Maintenance

List of [1] TH250A/[2] TH350A replacement parts for maintenance

Part name	Model	Type	Unit code	Maker	Q'ty	Remarks
AC servo motor	[1] [2]	MSMD022S2U	Y610A34D0	SHIBAURA MACHINE CO.,LTD.	1	Axis 1
	[1] [2]	MSMD012S2U	Y610A34E0		1	Axis 2
	[1] [2]	MSMD012S2R	Y610A37A0		1	Axis 3
	[1] [2]	MQMD012S2C	Y610A37B0		1	Axis 4
Reduction gear	[1]	M074824	Y610A3020		1	Axis 1
	[2]	M086369	Y610A3030		1	Axis 1
	[1] [2]	M074825	Y610A3040		1	Axis 2
Battery	[1] [2]	Size AA alkali battery (LR6AG)			3	
Planetary reduction gear	[1] [2]	M074826	Y610A3050		1	Axis 4
Timing belt	[1] [2]	S777403			1	Axis 3
	[1] [2]	S777404			1	Axis 4
Timing pulley	[1] [2]	S777399			1	Axis 3
	[1] [2]	S777400			1	Axis 3
	[1] [2]	S777401			1	Axis 4
	[1] [2]	S777402			1	Axis 4
Ball screw spline	[1] [2]	BNS1616AEFM+ 300LC5F	Y610A3010		1	Axis 3
Main harness	[1]	H140510G01	Y610A3740		1	
	[2]	H140510G02	Y610A3750		1	
Grease	[1] [2]	4B No. 2		Harmonic Drive Systems	500 g 2.5 kg	
	[1] [2]	OT–20 2		Solvay Solexis	2.5 kg	

! CAUTION

- With the exception of the grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts other than the batteries by the user.

List of TH180 replacement parts for maintenance

Part name	Type	Unit code	Maker	Q'ty	Remarks
AC servo motor	TS4606N2049E200	Y610A3620	SHIBAURA MACHINE CO.,LTD.	1	Axis 1
	TS4602N2049E200	Y610A3600		1	Axis 2
	TS4602N7049E200	Y610A3610		1	Axis 3
	TS4602N2049E200	Y610A3600		1	Axis 4
Reduction gear	S777267	Y610A3630		1	Axis 1
	S777268	Y610A3640		1	Axis 2
Battery	Size AA alkali battery (LR6AG)			3	
Planetary reduction gear	S777269	Y610A3650		1	Axis 4
Timing belt	243-3GT-9			1	Axis 3
	250-5GT-15			1	Axis 4
Timing pulley	S777217			1	Axis 3
	S777216			1	Axis 3
	S777219			1	Axis 4
	S777218			1	Axis 4
Ball screw spline	BNS1616AEFM +300LC5F	Y610A3010		1	Axis 3
Main harness	H14039G	Y610A3690		1	
	OT-20			1	
Grease	TS4606N2049E200	Y610A3620	Solvay Solexis	2.5 kg	

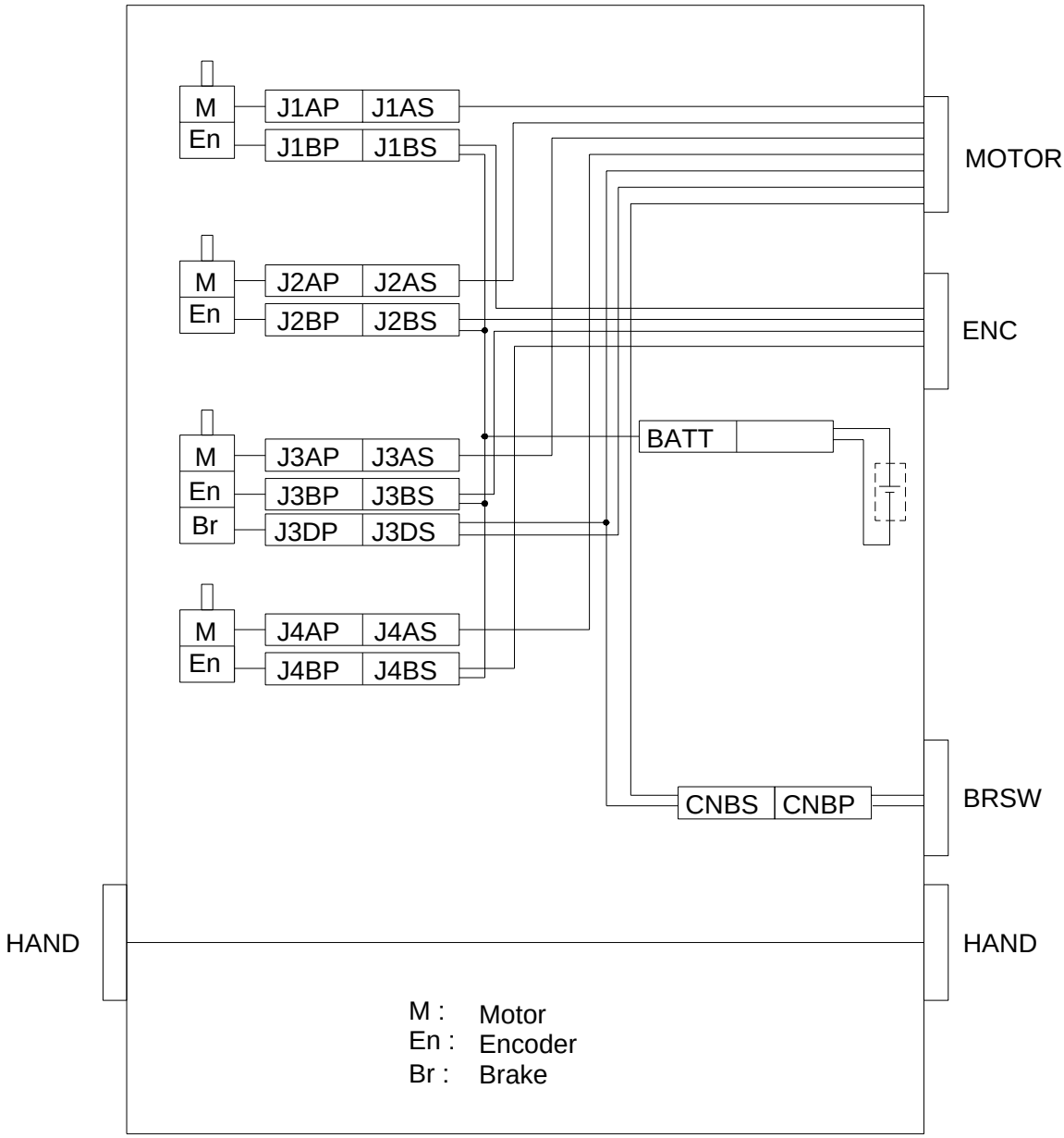
! CAUTION

- With the exception of the grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts other than the batteries by the user.

Appendix Table 2 List of Mechanical Replacement Parts for Maintenance

Part name	Type	Maker	Q'ty	Remarks
PS1 (Switching power supply)	LEB100F–0524	Cosel	1	
Main power switch	BAM205031	Panasonic Electronic Devices	1	
Lithium battery	ER6V–C	TOSHIBA LIFESTYLE PRODUCTS & SERVICES Corp.	1	
Fuse (X8LC printed board)	OMF–51NR 250 V 3A	World	1	
Printed circuit board	X8LC (Main control)	SHIBAURA MACHINE	1	
	X8LS (Servo)		1	
	X8LJ (Mother board also serving as connector board)		1	
	X8LO (Control panel)		1	
Teach pendant	TP1000	SHIBAURA MACHINE	1	with 5 m-long cable
IC	M54562P	Mitsubishi Electric	3	

Appendix Diagram 1 Component Connection Diagram



Appendix A Encoder Error

An encoder error will generate when replacement of the battery for position detector or the battery for X8LC printed board SRAM has not been performed regularly, or replacement of such batteries has been performed incorrectly. For the battery replacement, see Para. 2.3.8 and 3.3.3.
When the encoder error has generated, reset it, referring to Appendix B, Appendix C, Appendix D and Appendix E.
After the motor has been replaced or the home position has shifted, perform REORG operation followed by ZEROP operation. During the ZEROP operation, it is recommended to establish the home position, using the home point match-marks.
At other than the above, perform encoder error reset operation or multi-turn data reset operation. Also, when the multi-turn data error has generated at the same time with the encoder error, perform the multi-turn data reset operation.
When the encoder error cannot be reset only by the multi-data reset operation, perform the encoder error reset operation, then the multi-turn data reset operation.

Appendix B Encoder Error Reset Operation

This operation can be executed on the encoder status screen. Just after the encoder status screen has been called, the cursor points out the “Err-def” column on the “E1” line for the system that allows the encoder reset operation, and the “MULTI” column on the “E1” line for the system that prohibits the same operation.
The system that allows the encoder reset operation is TS1000 version 7.00 and after.
For the system that prohibits the encoder reset operation, execute the multi-turn data reset operation (Appendix C) to reset the encoder error.
When this operation is executed, no limitation is imposed on the robot position.

1) Press the “UTILITY” key on the teach pendant. Press the “NEXT” key until the [ENC] <F5> menu appears, then press the [ENC] <F5> key. Now, the encoder status screen is called on the display.

	MULTI	SINGLE	Err-def	ID
E1	0	00001234	**0200**	10
E2	0	00001234	0200	10
E3	0	00001234	4200	10
E4	0	00001234	0200	10
E5	0	00000000	0000	00
ERR		RESET		

2) Just after the encoder status screen has prevailed, the cursor is located at the “Err-def” column on the “E1” line. Move the cursor to a line where you wish to perform the encoder error reset operation, using the cursor keys (“↑” and “↓”).

	MULTI	SINGLE	Err-def	ID
E1	0	00001234	0200	10
E2	0	00001234	0200	10
E3	0	00001234	**4200**	10
E4	0	00001234	0200	10
E5	0	00000000	0000	00
ERR		RESET		

3) When you wish to reset the axis 3 encoder error, for instance, move the cursor to the “Err-def” column on the “E3” line, press the [RESET] <F3> key, then press the “EXE” key unless any problem arises. Thus, the axis 3 encoder error is reset. Call the normal error screen and press the [RESET] <F3> key to reset the error of “8–129 Axis3 Encoder abnormal”.

4) When the "←" cursor key is pressed twice in the condition of 1) above, the cursor moves to the "MULTI" column on the "E1" line. That is, the system enters the multi-turn data reset screen. For the operation under this condition, see Appendix C. In the system which allows the encoder error reset operation, when the "→" is pressed, the cursor currently pointing out the "MULTI" column moves to the "Err-def" column.
Under the "Err-def" column of the encoder status screen, the error status is expressed in hexadecimal notation. An example of the error status is given in Table 4.1 below. When it is displayed, you can perform the error reset operation, referring to Appendix B to Appendix E.
If a status other than those given in Table 4.1 is displayed under the "Err-def" column, contact us at the after-sale service department.

Table 4.1 Example of "Err-def"

Error status	Remarks
0000	Normal condition.
0100	This error occurs when heavy vibration has been exerted on the robot or the robot has been moved fast by hand while the motor drive line was disconnected at power OFF. The multi-turn data has possibly shifted.
0200	The encoder resolution has deteriorated. During the operation, the encoder resolution will be restored normally.
4000	The battery voltage has dropped. Replace the battery, and the voltage returns to the normal level.
8000	The battery voltage has dropped further than the above. The robot home position has possibly been lost, or the multi-turn data error has possibly generated at the same time.
C000	Error status "4000" has generated together with error status "8000".
4200	Error status "4000" has generated together with error status "0200".

Appendix C Multi-Turn Data Reset Operation

This operation should be executed after the home position of an appropriate axis has been established, using the home point match-marks, that is, after the relevant axis has returned to the home position.
As only the multi-turn data of the absolute encoder can be zero-reset and no single-turn data is affected by this operation, the axis can be restored to a status which is quite the same as the status prevailing before position error generation.
Also, as described above, the encoder error can be reset in lieu of the encoder error reset operation.

1) Press the “UTILITY” key on the teach pendant. Press the “NEXT” key until the [ENC] <F5> menu appears, then press the [ENC] <F5> key. Now, the encoder status screen is called on the display.

	MULTI	SINGLE	Err-def	ID
E1	**1234**	00001234	0200	10
E2	2345	00001234	4200	10
E3	3456	00001234	0200	10
E4	4567	00001234	0200	10
E5	0	00000000	0000	00
ERR		RESET		

2) Move all axes for which multi-turn data reset operation is to be executed to respective home positions. Minor error is acceptable.

3) The multi-turn data reset function is normally protected because it will seriously influence the robot current position by wrong operation.
To cancel this protect function, press the “ALT” and “0” keys at the same time.

4) Only after the protect mode is canceled, the cursor can be moved. Move the cursor to a line where you wish to perform the multi-turn data reset operation, using the cursor keys (“↑” and “↓”). In the system which does not allow the encoder error reset operation, the “→” cursor key is inoperative.

5) To reset the axis 2 multi-turn data, for instance, move the cursor to the “MULTI” column on the “E2” line.

	MULTI	SINGLE	Err-def	ID
E1	1234	00001234	0200	10
E2	**2345**	00001234	4200	10
E3	3456	00001234	0200	10
E4	4567	00001234	0200	10
E5	0	00000000	0000	00
ERR		RESET		

6) Press the [RESET] <F3> key and press the “EXE” key if everything is OK.

	MULTI	SINGLE	Err-def	ID
E1	1234	00001234	0200	10
E2	**0000**	00001234	4200	10
E3	3456	00001234	0200	10
E4	4567	00001234	0200	10
E5	0	00000000	0000	00
ERR		RESET		

7) Now, the axis 2 multi-turn data is reset to zero (0) at the same time that the encoder error is reset. Call the normal error screen and press the [RESET] <F3> key to reset the error of “8–129 Axis2 Encoder abnormal”.

Appendix D REORG Operation

This operation should be performed only after each axis has been moved to the mechanical stopper. Then perform the ZEROP operation.
In this operation, minor error may be caused according to the adjustment method.

(1) Outline of HOME function

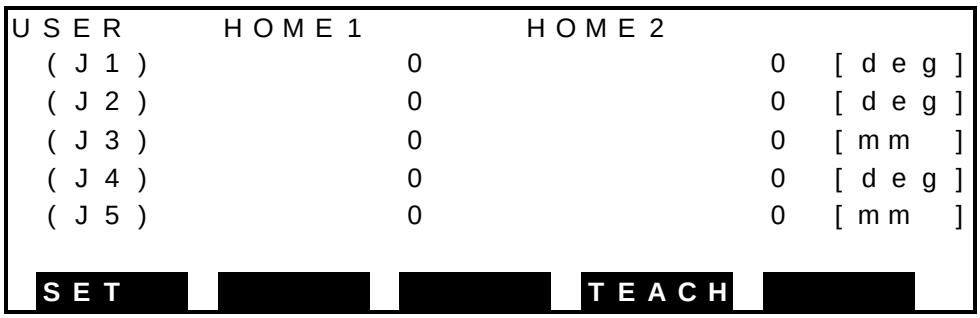
This function memorizes a specific coordinate value in each axis and restores the original home position data by using the HOME function when the machine home position data has been destroyed.

The set points are provided for HOME1 to HOME4, and the axis 1 to axis 3 plus (+) side mechanical stopper positions are set in HOME3, and the axis 1 to axis 3 minus (–) mechanical stopper positions in HOME4.

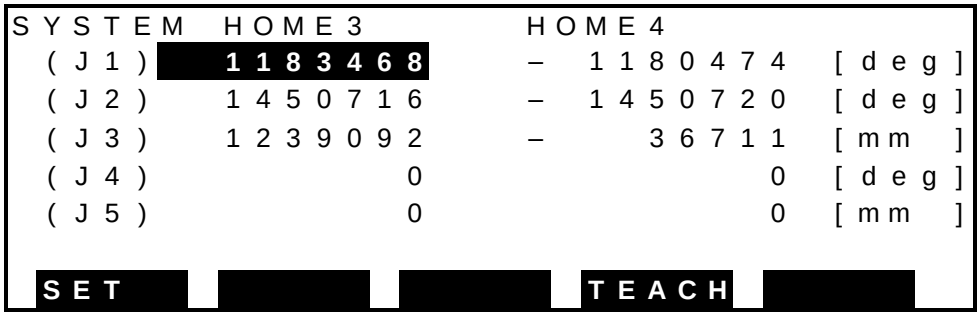
For HOME1 and HOME2, you can specify any values. It is recommended that when building up a system, you specify the place and clamp for securing the robot end.

On the following REORG screen, the unit of each axis position is "$\times 10^{-4}$ [deg], [mm]". For example, the value of "(J1) 1183468" below signifies that the axis 1 is positioned at 118.3468 [deg].

(Page 1)



(Page 2)



Caution: This function is used to restore the machine home point of each axis. NEVER use the function at other than the setting for restoration.

(2) Operating procedures

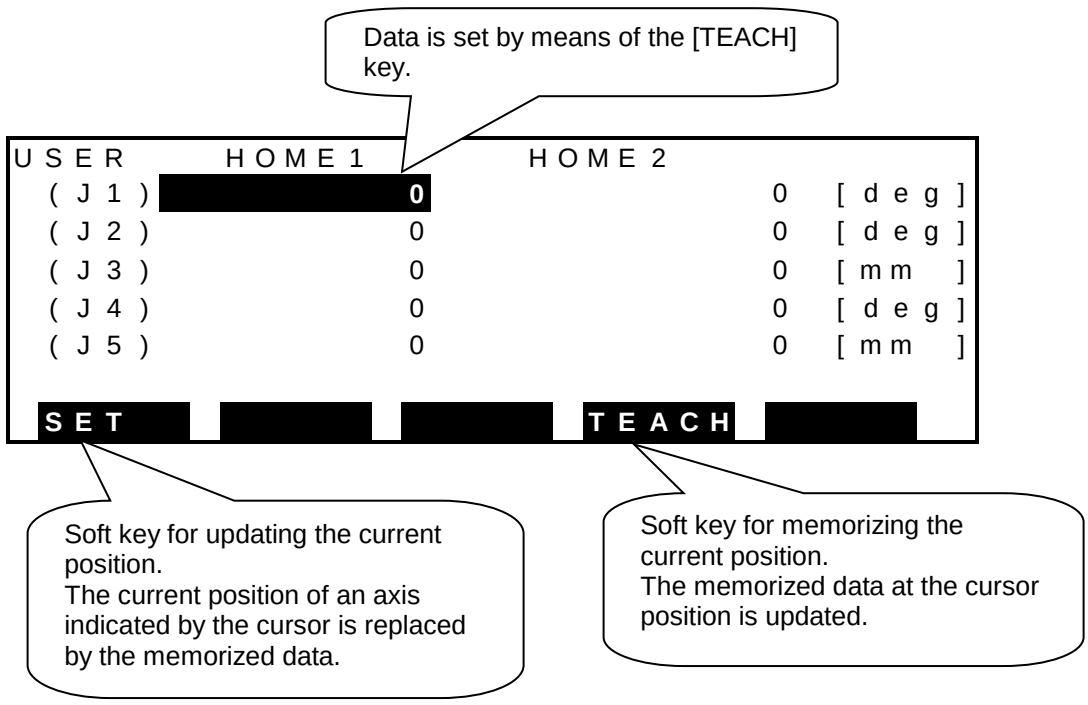
I. HOME setting:
In this paragraph, setting of values in HOME1 is explained. (The same is also applicable to HOME2.)

[1] Guide the robot to secure the ball screw spline shaft at a desired position.

[2] Turn off the servo power.

[3] Press the “UTILITY” key equipped on the teach pendant.

[4] Press the “NEXT” key until the [REORG] menu appears, then press the [REORG] key.



[5] When the above screen appears, memorize the position where the robot was secured for each axis (J1 ~ J4) by moving the cursor and using the [TEACH] mode.

[6] To establish the above, press the “EXE” key.

II. How to restore the data by HOME1:
If coordinate data of he machine home point has been destroyed, the machine home point can be restored, using the HOME1 function above.

[1] Guide the robot and secure it at the position set in HOME1 above, by using the clamp, etc.

[2] Turn off the servo power.

[3] Press the “UTILITY” key equipped on the teach pendant.

[4] Press the “NEXT” key until the [REORG] menu appears, then press the [REORG] key.

[5] Move the cursor to each axis under HOME1 of the REORG screen, then press the [SET] key to update the coordinate value set in HOME1 to the position where the robot was secured. Thus, the machine home position can be restored. At this time, before setting the axis 3, set the axis 4.

[6] When everything is OK, press the “EXE” key.

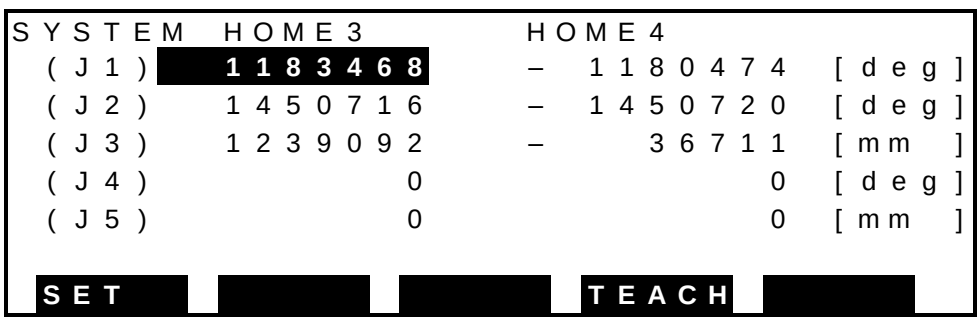
III. How to restore the data by HOME3 (or HOME4):
The plus (+) side mechanical stopper positions and minus (–) side mechanical stopper positions of the axes 1 to 3 are factory-set in HOME3 and HOME4, respectively. The machine home point can be restored, using the above.
This paragraph deals with the restoration procedures, using HOME3.

[1] Turn off the servo power.

[2] Press the “UTILITY” key provided on the teach pendant.

[3] Press the “NEXT” key until the [REORG] menu appears, then press the [REORG] key.

[4] Press the “NEXT” key to call Page 2.



[5] Contact the axis 1 of the robot to the plus (+) side mechanical stopper by hand.

[6] Move the cursor to (J1) of HOME3, then press the [SET] key. To establish, press the “EXE” key. Now, the machine home point of the axis 1 is restored.

[7] Likewise, contact the axis 2 of the robot to the plus (+) side mechanical stopper by hand.

[8] Move the cursor to (J2) of HOME3, then press the [SET] key. To establish, press the “EXE” key. Now, the machine home point of the axis 2 is restored.

[9] Press the [ESC] key to escape from the REORG screen once.

[10] Turn on the servo power, and set the axes 3 and 4 in the servo-free state.

[11] Set the axis 4 to the home point match-mark.

[12] Contact the axis 3 of the robot to the plus (+) side mechanical stopper by hand.

[13] Turn off the servo power.

[14] Repeat Steps [2] to [4] inclusive to call the HOME screen again.

[15] Move the cursor to (J4) of HOME3, then press the [SET] key. To establish, press the “EXE” key. Now, the machine home point of the axis 4 is restored.

[16] Move the cursor to (J3) of HOME3, then press the [SET] key. To establish, press the “EXE” key. Now, the machine home point of the axis 3 is restored.

[17] Press the [ESC] key to escape from the HOME screen. Now, the operation has finished.

Cautions:

- Be sure to set the home point in the order of E4 $\rightarrow$ E3 because E3 (axis 3) and E4 (axis 4) of the SCARA robot are interlocked with each other.
- If the axis 1, 2 or 3 hits each mechanical stopper, an alarm will generate. To proceed to the next operation, cancel the alarm. This alarm occurs when the mechanical stopper exceeds the soft limit position.

Appendix E ZEROP Operation

This operation is the home point setting operation which is executed in the maker's plant before shipment. Execute the operation by using the exclusive fixtures.
This restoring method may also cause a minor error according to the adjustment.

a) Turn on the controller power and align the home point match-marks of each axis in the manual mode.
For the axis 4, however, align the center of the ball screw spline groove (left) with the axis 4 home point match-mark, as shown in Fig. 4.2.
If this is not possible, mark the reference position of each axis and jot down its coordinate beforehand. Or it is recommended that when building up the system, you specify the place and clamp for securing the robot end.

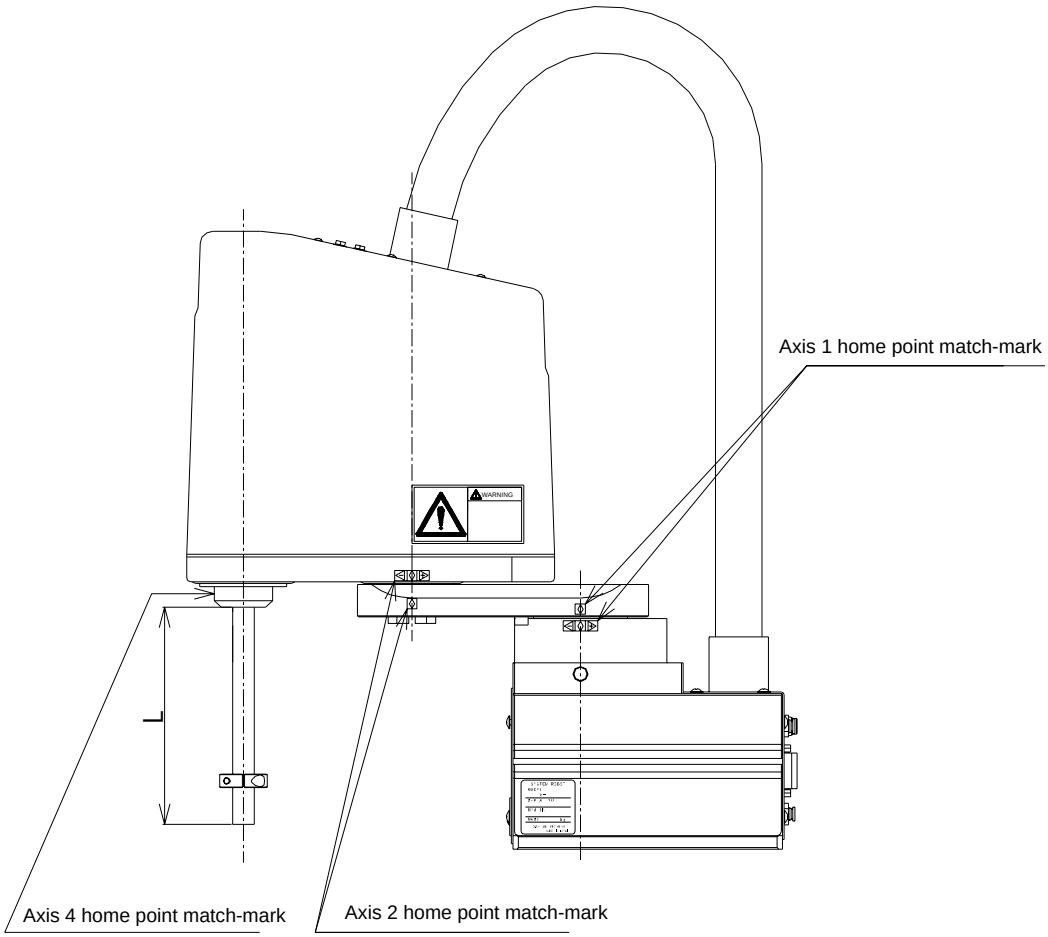


Fig. 4.1 Locations of home point match-marks of respective axes

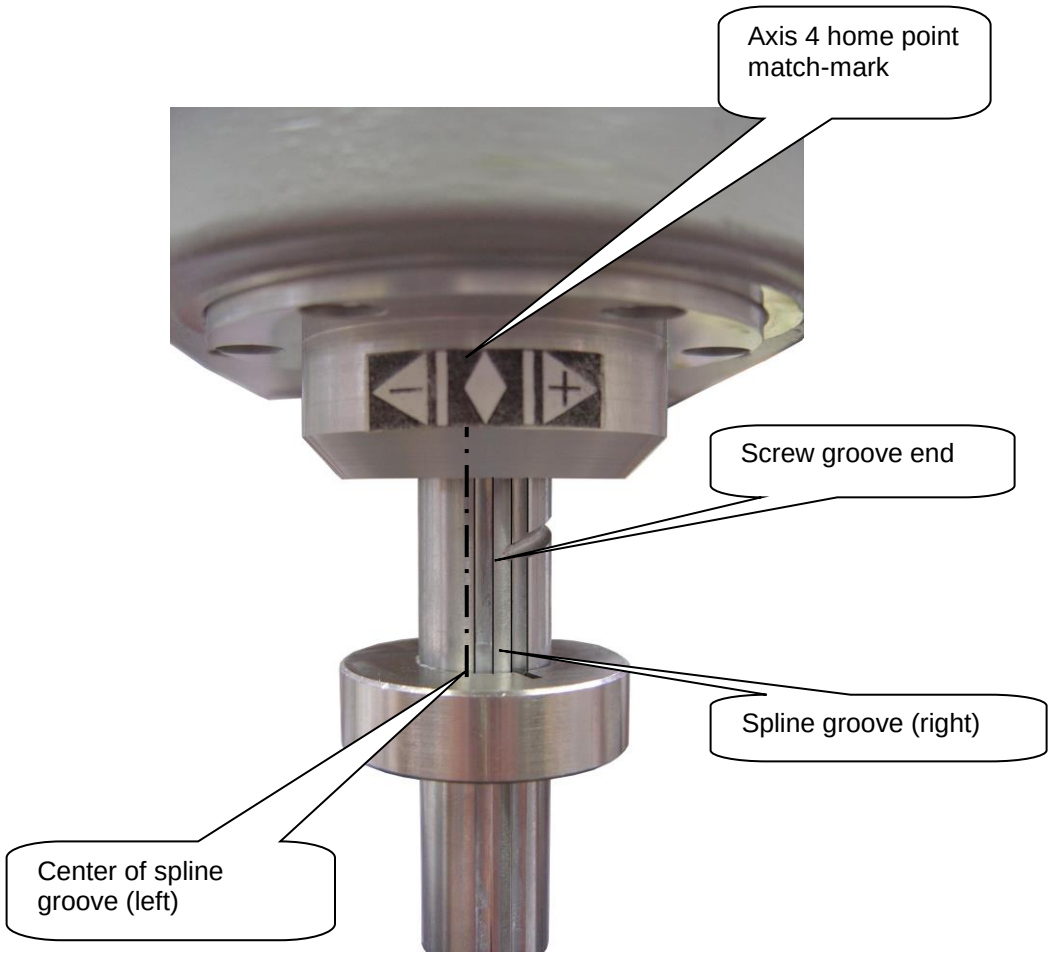


Fig. 4.2 Axis 4 home point match-mark position and spline groove (when viewed from the front side)

b) Because no home point match marks are provided for the axis 3, establish the axis 3 home point in the following manner.
Move the axis 3 down to near the lower limit and set length “L” in Fig. 4.1 above (i.e., distance between the boss end surface of the ball screw spline and hand flange end surface) to the following value. This position is the axis 3 home position (Z = 0).

TH250A : 160.5 mm
TH350A : 160.5 mm
TH180 : 162 mm

c) Setting of new home position data (To be set through teach pendant.)

1) Select the TEACHING mode by means of the master mode switch.

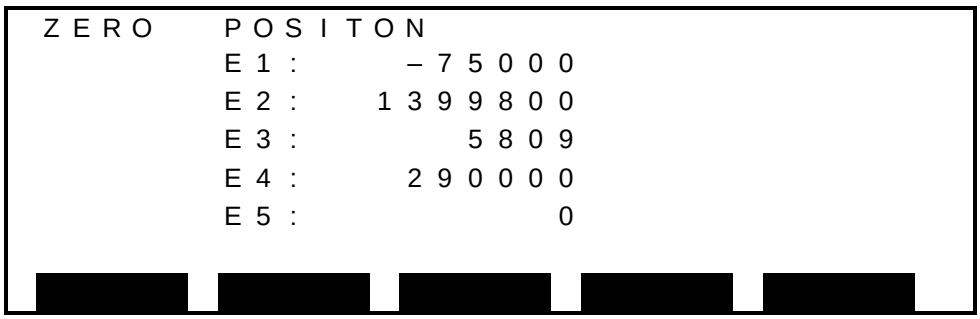
2) Press the “UTILITY” key equipped on the teach pendant.

3) Press the “NEXT” key on the teach pendant twice.

4) Turn off the servo power.

5) Press the [F3] key on the teach pendant to select the ZEROP mode.

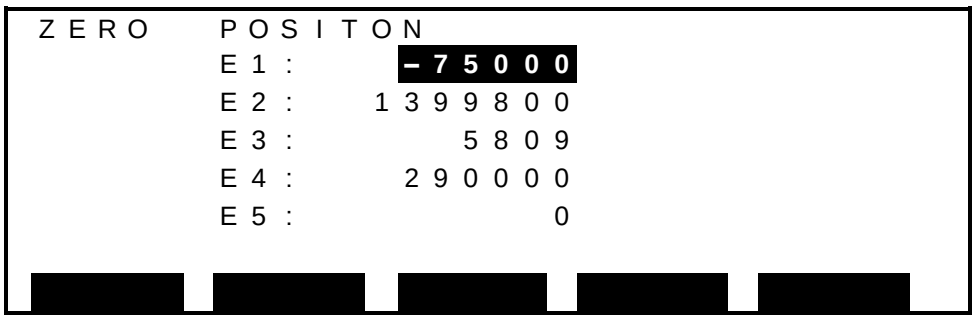
6) The following home point setting screen is called.
The unit of each axis position is “$\times 10^{-4}$ [deg], [mm]”. For example, the value of “E1 –75000” below signifies that the axis 1 is positioned at –7.5000 [deg].



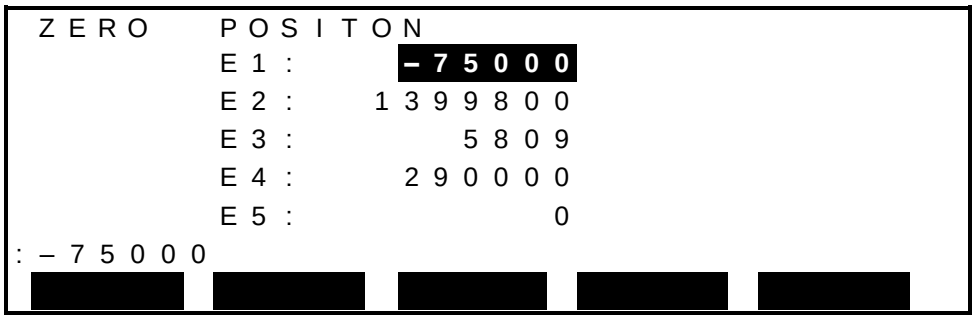
7) Press the “ALT” and “0” keys on the teach pendant.

8) The following home point editing screen appears.

9) The numerical value of E1 (axis 1) is inversely displayed.

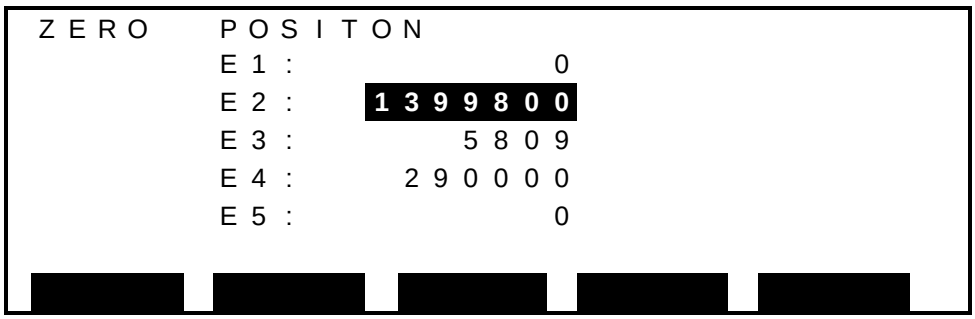


10) Press the “EXE” key on the teach pendant, and the value of “E1” is displayed at the lower left side. Set “0” in lieu of this value and press the “EXE” key. Then the E1 value becomes “0”.
Unless the axis could be located at the home point mark, enter the coordinate value registered beforehand.



11) When the “↓” key on the teach pendant is pressed, the following E2 value is displayed by reversed image. Perform setting in the same manner as above.

Caution: Be sure to set the home point in the order of E4 → E3 because E3 (axis 3) and E4 (axis 4) of the SCARA robot are interlocked with each other.



d) Confirmation of home position

1) Make sure of the robot current position according to the POS data of the UTILITY mode. Guide the robot to the approximate home position by manual operation and confirm the position. Unless it could be located at the home point mark, make sure that the approximate position is the reference position marked in advance.

2) Execute the following program and make sure of the robot machine home position. Unless the robot could be located at the home point mark, DO NOT use the following program.
Create a program in which the robot moves to the position recorded beforehand, referring to the following program. Then make sure that the robot moves to the reference position mark according to the program thus created.

```
PROGRAM ZERO
 SPEED=30
 MOVEA 3,100
 MOVEA 1,0
 MOVEA 2,0
 MOVEA 3,0
 MOVEA 4,0
END
```

e) Data saving
Save the “MACHINE. PAR” file stored in the controller to another media.
Connect the personal computer and controller using TSPC (option), insert the system disk into the floppy disk unit and save the “MACHINE.PAR” file from the robot controller to the personal computer.
The home position data is the data characteristic of the robot, which differs with the robot you use. This data is necessary when you effect a cold start and restore the home position data again.

f) Now, the operation completes.